

Exam Subject: Applied Deep Learning
 Appointment: XX.XX.XXXX
 Permitted Aids: **One Din A4 page handwritten**
 Examiner: Prof. Dr. Christian Groth

Matriculation number:

Important notes:

- Please check the assignment sheets for completeness:
 - 1 Cover page
 - XX task pages
 so a total of XX pages.
- Write the solutions directly after the task on the respective sheets. If you notice during the processing that there is not enough space on the task sheets, use the back pages and write the corresponding solutions on them with a fully qualified indication of the sub-task processed in each case (e.g. "Task 2, Sub-task 1a").
- This sheet, as well as all assignment sheets or other sheets related to the assignments, must be marked with your **matriculation number**.
- Tasks or parts of tasks are only considered solved if the individual solution steps are legible and understandable. Only one solution may be submitted per task.

To be filled in by the examiner

Interruptions:

from _____ to _____
 from _____ to _____

Comments:

(grade)	Signature of first examiner	(grade)	Signature second examiner

Matriculation number:

1 Basics

Just answer with yes / no.

Statement	Yes	No
Can a model with a very low training error still have poor generalization performance?		
In a Recurrent Neural Network (RNN), are weights shared across time steps?		
Is the softmax function generally more suitable for regression tasks than for classification?		
Are CNNs used to classify texts?		
A high number of parameters increases the accuracy of the model.		
Dropout is a method of regularization.		
In transfer learning we can just replace the last layer of an existing neural network.		
Sigmoid function will always return values between 0 and 1.		
Should categorical variables always be converted into numerical form before being used in neural networks?		
An epoch is when the model has processed all the data.		

2 Data preparation

- a) Can removing outliers before splitting into train and test sets introduce bias in model evaluation? Briefly explain your decision.

- b) Why is it important to scale features before using gradient-based learning algorithms?

- c) Explain the difference between standardization and normalization.

- d) How can you deal with text data (e.g. free-text fields) before feeding it into a neural network? Name two possibilities.

- e) Name two strategies for handling outliers during preprocessing and explain when you might use them.

- f) Give one example of a categorical variable.

- g) What does a train-test split help you measure?

- h) Your dataset includes timestamps, and you want to use them to predict customer purchase behavior. A colleague suggests dropping them because they are "not useful."
Explain why this might be a mistake. Suggest at least one derived feature you could create from the timestamp and how it might help the model.

- i) When do you drop a column or a row in data preparation?

3 Machine Learning

- a) Explain the terms bias, variance, and bias-variance trade-off.

- b) Which activation function do you use in the last layer of an NN for multi-class classification?

- c) What is momentum at SGD? In what way is it helpful?

- d) What is batch normalization?

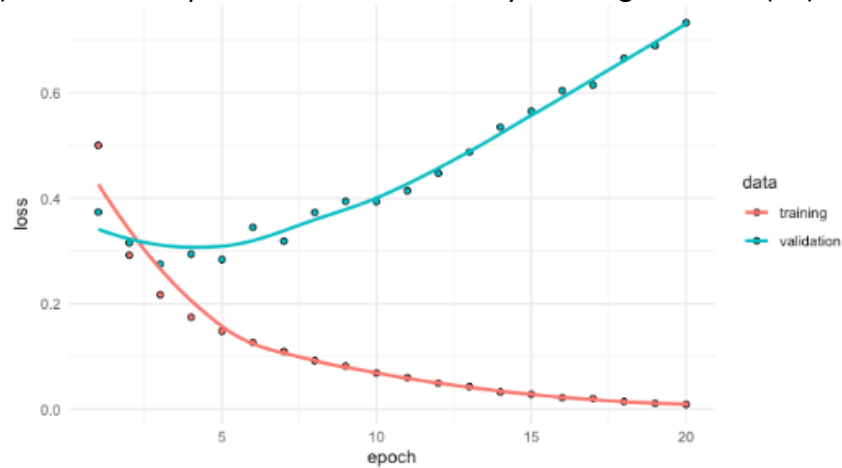
- e) Describe a Possible Data - Imputation Technique.

Matriculation number:

f) Why are CNNs better for processing image data?

g) What is k-fold cross validation?

h) What is the problem here? How can you recognize this? (2P)



i) How many parameters need to be trained when we apply a 3x3x3 filter to a 32x32x3 image in a CNN? (2P)

- a) Why can deeper neural networks model more complex functions than shallow ones, and what are the limitations of simply adding more layers?
- b) Describe the purpose of using activation functions like ReLU, Leaky ReLU, or GELU in deep networks.
- c) You attempt to train a deep CNN from scratch on a small dataset of satellite images. The loss becomes NaN after a few iterations. You double-checked the labels and input scaling. List two causes related to deep architectures that can lead to unstable or undefined loss values. Explain why they occur and how to mitigate them.
- d) What is dropout? How does it work?

2 Transfer Learning

- a) What is the difference between a classical ML approach and an approach with transfer learning?

- b) You are building a deep learning model to classify plant leaf diseases from photographs taken in the field. However, you only have a dataset of 1,200 labeled images. A colleague suggests using a CNN pre-trained on ImageNet.

What are the advantages of using a pre-trained model in this context?

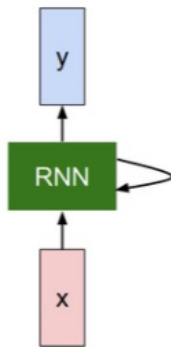
Which layers would you likely freeze or fine-tune, and why?

What would change if your images were taken using a thermal camera instead of RGB?

3 RNNs

a) Describe the problems of MLPs and CNNs in processing time series data.

b) Sketch an unrolling of the following mesh for 3 consecutive inputs x_1, x_2, x_3 .



c) What is meant by vanishing gradient?

Efficient training

- a) What is a learning rate scheduler and why do we need it?
- b) How does L1 regularization prevent overfitting?
- c) Why are GPUs better suited for the training of deep neural nets than CPUs?
- d) What is model sharding and when do we need it?

5 Reinforcement Learning

- a) Explain the difference between a rational agent and an omniscient agent.
- b) Give one example each of a dynamic and static environment, including the agent's task.

Matriculation number:

- c) What is a table-driven agent and what are its drawbacks?

- d) Explain the Exploitation Exploration Trade-off

- e) What does the input and output layer look like in a deep q learning neural network?

- f) What is the effect of learning rate α in the context of Q-learning?

6 Code Understanding

Consider the following source code snippet:

```
class DataModel(pl.LightningModule):
    def __init__(self):
        super(DataModel, self).__init__()
        self.loss = nn.MSELoss()
        self.layers = nn.Sequential(nn.Flatten(),
                                    nn.Linear(784, 64),
                                    nn.ReLU(),
                                    nn.Dropout(),
                                    nn.Linear(64, 64),
                                    nn.ReLU(),
                                    nn.Dropout(),
                                    nn.Linear(64, 10))

    def configure_optimizers(self):
        params = self.parameters()
        optimizer = optim.SGD(params=params, lr = 0.01)
        return optimizer
```

a) How many layers does this neural network have?

b) How many input parameters?

c) What is the learning rate?

d) What kind of loss function is used?